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import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, r2_score

# تحميل البيانات
df = pd.read_csv('egypt_tourism_preprocessed.xls')

features = ['age', 'gender', 'nationality', 'city_visited', 'trip_duration_days', 'season']

df = df.dropna(subset=features + ['total_spending_usd', 'spending_per_day_usd'])

X = df[features]
X = pd.get_dummies(X, columns=['gender', 'nationality', 'city_visited', 'season', 'travel_

# ----- الموديل الأول: التنبؤ بإجمالي صرف الرحلة كلها -----
y_total = df['total_spending_usd']
X_train, X_test, y_train, y_test = train_test_split(X, y_total, test_size=0.2, random_sta

lr_total = LinearRegression()
lr_total.fit(X_train, y_train)
pred_lr_total = lr_total.predict(X_test)

rf_total = RandomForestRegressor(n_estimators=100, random_state=42)
rf_total.fit(X_train, y_train)
pred_rf_total = rf_total.predict(X_test)

print("==== Total Trip Spending ====")
print("Linear Regression - MAE:", mean_absolute_error(y_test, pred_lr_total), "R2:", r2_s
print("Random Forest      - MAE:", mean_absolute_error(y_test, pred_rf_total), "R2:", r2_s

# ----- الموديل الثاني: التنبؤ بصرف اليوم الواحد -----
y_daily = df['spending_per_day_usd']
X_train2, X_test2, y_train2, y_test2 = train_test_split(X, y_daily, test_size=0.2, random

lr_daily = LinearRegression()
lr_daily.fit(X_train2, y_train2)
pred_lr_daily = lr_daily.predict(X_test2)

rf_daily = RandomForestRegressor(n_estimators=100, random_state=42)
rf_daily.fit(X_train2, y_train2)
pred_rf_daily = rf_daily.predict(X_test2)

print("\n==== Daily Spending ====")
print("Linear Regression - MAE:", mean_absolute_error(y_test2, pred_lr_daily), "R2:", r2_
print("Random Forest      - MAE:", mean_absolute_error(y_test2, pred_rf_daily), "R2:", r2_

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==== Total Trip Spending ====
Linear Regression - MAE: 260.02329434265766 R2: 0.7412555801222859
Random Forest      - MAE: 249.32084721833334 R2: 0.7334110043782559

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==== Daily Spending ====
Linear Regression - MAE: 39.00041087740218 R2: 0.4578522087006992
Random Forest      - MAE: 39.96835727166667 R2: 0.41036136455591987

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import pandas as pd
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# Load data (upload the file to Colab first)
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# Load data (upload the files to Colab first)
tourism_df = pd.read_csv('egypt_tourism_preprocessed.xls')
hotels_df = pd.read_csv('Egypt_hotels_preprocessed.xls')

# ----- Tourist statistics -----
print("==== Average Spending & Duration =====")
print("Average daily spending (USD):", tourism_df['spending_per_day_usd'].mean())
print("Average total trip spending (USD):", tourism_df['total_spending_usd'].mean())
print("Average trip duration (days):", tourism_df['trip_duration_days'].mean())

print("\n==== Most Visited Cities =====")
print(tourism_df['city_visited'].value_counts())

print("\n==== Most Used Travel Companies =====")
print(tourism_df['travel_company'].value_counts())

print("\n==== Average Spending by City =====")
print(tourism_df.groupby('city_visited')['total_spending_usd'].mean().sort_values(ascending=

print("\n==== Average Spending by Season =====")
print(tourism_df.groupby('season')['total_spending_usd'].mean().sort_values(ascending=False)

print("\n==== Average Spending by Nationality =====")
print(tourism_df.groupby('nationality')['total_spending_usd'].mean().sort_values(ascending=F

# ----- Hotel statistics -----
print("\n==== Top 10 Rated Hotels =====")
top Rated = hotels_df.sort_values('overall_rating', ascending=False).head(10)
print(top Rated[['name', 'overall_rating', 'reviews']])

print("\n==== Top 10 Most Reviewed Hotels =====")
most Reviewed = hotels_df.sort_values('reviews', ascending=False).head(10)
print(most Reviewed[['name', 'overall_rating', 'reviews']])

print("\n==== Overall Average Hotel Ratings =====")
print("Average overall_rating:", hotels_df['overall_rating'].mean())
print("Average location_rating:", hotels_df['location_rating'].mean())

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440	Luca Pyramids Inn	5.0
466	KEMET GUEST HOUSE	5.0

	reviews
536	2
533	1
543	8
539	21
607	5
662	23
487	1
490	3
446	1
466	2

===== Top 10 Most Reviewed Hotels =====

	name	overall_rating	reviews
252	Rixos Premium Seagate	4.9	44350
149	SUNRISE Royal Makadi Resort	4.9	36488
134	Rixos Sharm El Sheikh	4.9	36285
123	Rixos Sharm El Sheikh	4.9	36285
91	Sofitel Cairo Nile El Gezirah	4.8	34244
23	Ramses Hilton	4.2	26537
84	InterContinental Citystars Cairo, an IHG Hotel	4.7	24871
159	InterContinental Citystars Cairo, an IHG Hotel	4.7	24871
164	SUNRISE Arabian Beach Resort -Grand Select-	4.8	24436
17	Novotel Cairo Airport	4.6	22083

===== Overall Average Hotel Ratings =====

Average overall_rating: 4.300399979806139

Average location_rating: 3.6051983584131326